

The target parameter

A confidence interval (CI) estimates a population parameter. Estimating a population mean μ from sample data is inferential statistics.

For transfer duration, does μ describe one transfer, the sample mean, or the population mean?

A point estimate

A point estimate uses one sample statistic to estimate a population parameter.

Which statistic estimates the population mean transfer duration?

An interval estimate

An interval estimate gives a range of plausible values for a population parameter.

Why might an interval communicate more uncertainty than a single sample mean?

A confidence interval form

A two-sided mean interval is centered at the sample mean.

$$\bar{X} \pm EBM$$

If the mean is 12 and the margin of error is 1.5, what are its endpoints?

Margin of error

The margin of error, or error bound for a population mean (EBM), is the distance from the point estimate to either endpoint.

For an interval from 10 to 14 minutes, what is the margin of error?

Confidence level

The confidence level (CL) is the long-run capture rate. A valid 95% procedure captures the fixed parameter in about 95% of repeated intervals.

What would 95% describe if we repeatedly sampled and constructed intervals?

A repeated-sample view

Imagine 100 independent random samples, each producing a 95% confidence interval using a valid procedure.

Should exactly 95 of the intervals be guaranteed to contain μ ?

The two tails

Alpha (α) is the long-run noncoverage probability, $1 - CL$. A central normal interval divides it equally between two tails.

$$\alpha = 1 - CL$$

For 95% confidence, how much area belongs in each tail?

A critical z value

The positive critical value z^* , written $z_{(\alpha/2)}$ in OpenStax, cuts off $\alpha/2$ in the upper standard-normal tail.

$$P(Z > z^*) = \frac{\alpha}{2}$$

Which area to the left identifies z^* for 95% confidence?

Finding the critical value

For 95% confidence, the standard-normal critical value is approximately 1.96.

What probability lies between -1.96 and 1.96?

The known-SD method

A z interval for μ uses a known population SD and a normal or adequately approximated sampling distribution of the mean.

What would you check beyond sample size before choosing this method?

Standard error for the interval

Synthetic independent random sample: $n=64$, mean=12 minutes. The population is normal with known SD 4 minutes.

What standard error belongs in the confidence interval?

The mean margin of error

For a known population SD, the mean margin of error is a critical value times standard error.

$$EBM = z^* \frac{\sigma}{\sqrt{n}}$$

For $\sigma=4$, $n=64$, and $z^*=1.96$, what is EBM?

Constructing a mean interval

Synthetic independent random sample: $n=64$, mean=12 minutes. Normal population; known SD=4 minutes; 95% confidence; $z^*=1.96$.

What is the confidence interval for μ ?

A mean interval from raw data

Synthetic independent normal sample (min): 9, 11, 12, 14. Known population SD = 2; 90% confidence; $z^* = 1.645$.

What confidence interval for μ follows from the raw observations?

Interpreting the interval

Synthetic result: a 95% confidence interval for mean transfer duration is 11.02 to 12.98 minutes.

How would you state this result about the population mean?

A fixed parameter

Synthetic result: one calculated 95% interval is 11.02 to 12.98 minutes. The population mean is fixed.

Does the procedure assign a 95% probability to that fixed mean lying inside this particular interval?

Mean versus individual values

Synthetic result: a 95% confidence interval for μ is 11.02 to 12.98 minutes.

Does this imply that 95% of individual transfer durations lie in the interval?

Recovering the point estimate

Synthetic result: a symmetric mean confidence interval is 18.4 to 21.6 seconds.

What sample mean produced this interval?

Recovering the margin

Synthetic result: a symmetric mean confidence interval is 18.4 to 21.6 seconds.

What is its margin of error?

More confidence

Use the same sample and known population SD. Compare 90% and 99% confidence intervals.

Which interval is wider?

More observations

Model: population SD and confidence level remain fixed. Increase n from 64 to 256.

How does the margin of error change?

More population variation

Keep sample size and confidence level fixed.
Double the known population SD.

How does the interval width change?

Planning a sample size

For known σ , a desired margin E determines a sample size, rounded upward.

$$n = \left(\frac{z^* \sigma}{E} \right)^2$$

For $\sigma=4$, $E=0.5$, and $z^*=1.96$, what n is needed?

A precision target

Model: a 95% mean interval has margin 1 minute at $n=64$, with known population SD.

What sample size would target a margin of 0.5 minutes?

Precision and bias

A synthetic sample measures only successful transfers; failed transfers are excluded.

Would increasing n make its confidence interval valid for all transfer attempts?

A value inside the interval

Synthetic result: a 95% interval for μ is 11.02 to 12.98 minutes. A claim states $\mu=12$.

Is the claimed value compatible with this interval?

Sketching the confidence interval

Synthetic known- σ result: a 95% confidence interval for μ is (18, 22), with sample mean 20.

How would you mark the estimate, endpoints, and error bound, and show the critical tail areas on a z curve?